BRAC UNIVERSITY
Department of Computer Science and Engineering

Examination: Surprise Test - 4
Duration: 10 minutes

Semester: Spring 2025
Full Marks: 05

CSE 340: Computer Architecture

Name: Solution	ID:	Section:

1. Write the following control bits for "LD X1, 16(X15) " [0.5*4 = 2]

Control	Value
RegWrite	1
MemRead	1
MemWrite	0
Branch	0

2. **Identify** the necessary resources or components from the list provided to construct the datapath for each of the following instructions. Just put a tick(✔) sign in the respective cells to show your selection.[3]

		PC	Instruction Memory	Data Memory	Register File	ALU	Immediate Generation Unit	Control Unit	ALU Control
i	ADD X21, X22, X23	✓	✓		✓	✓		✓	✓
ii	Slli X21, X22, 23	✓	✓		✓	✓	✓	✓	✓
iii	SB X21, 10(X22)	✓	✓	✓	✓	✓	✓	✓	✓